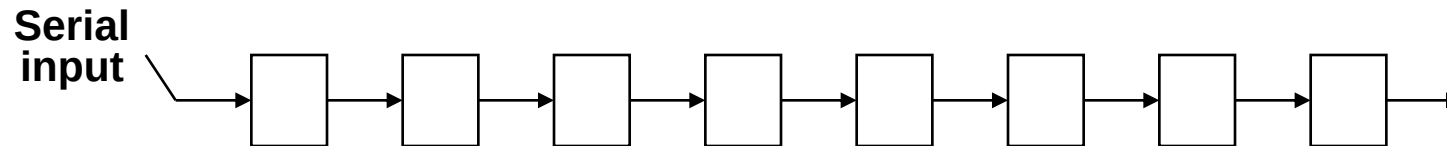


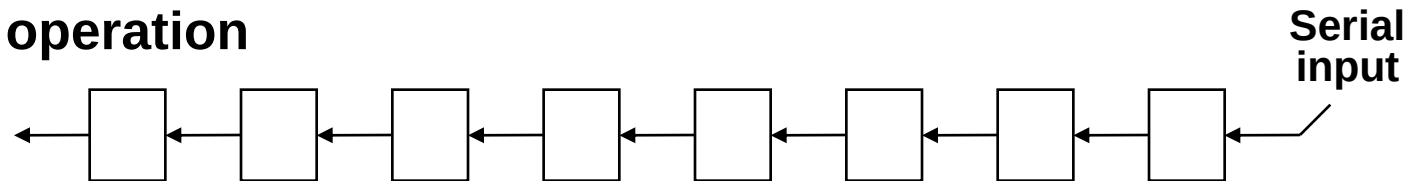
SHIFT MICROOPERATIONS/ Shifter circuit

- There are three types of shifts
 - *Logical shift*
 - *Circular shift (Barrel Shifter)*
 - *Arithmetic shift*
- What differentiates them is the information that goes into the serial input

- **A right shift operation**

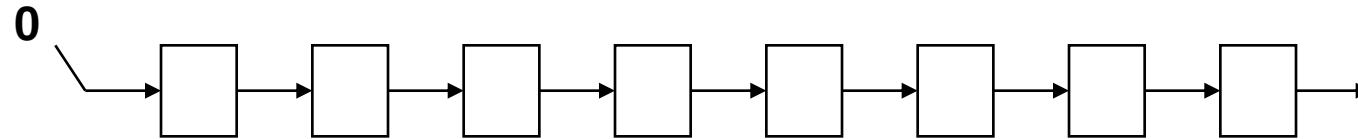


- **A left shift operation**

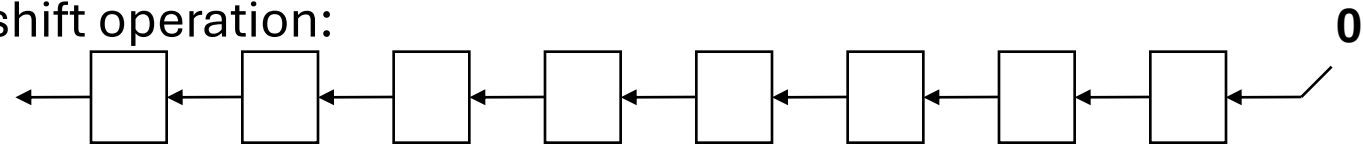


LOGICAL SHIFT

- It shift the number to the left or right and fills empty spots with 0's.
- A right logical shift operation:



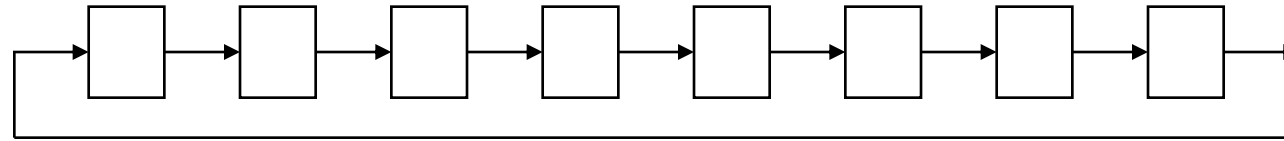
- A left logical shift operation:



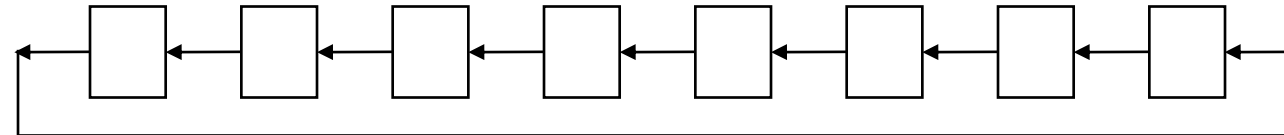
- Logical shift left 2
- Examples:
 $110110 \rightarrow 0110_ \rightarrow 011000$
- Logical shift right 1
- Examples:
 $110110 \rightarrow _11011 \rightarrow 011011$

CIRCULAR SHIFT/ Barrel shifter

- In a circular shift the serial input is the bit that is shifted out of the other end of the register.
- A right circular shift operation:



- A left circular shift operation:



Circular shift left 2

Examples:

110110 $\rightarrow$ 0110__ $\rightarrow$ 011011

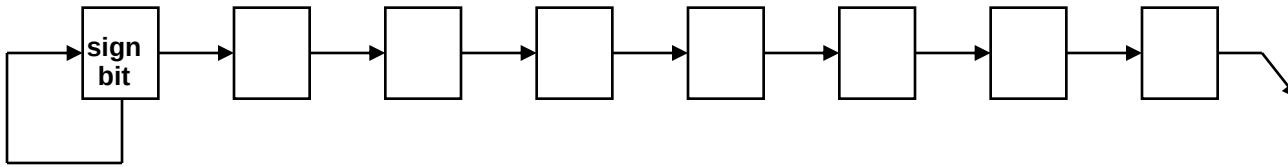
Circular shift right 1

Examples:

110110 $\rightarrow$ _ 11011 $\rightarrow$ 011011

ARITHMETIC SHIFT

- Same as logical shifter, but on right shift fills the MSB with copies of the sign bit.
- A right arithmetic shift operation:

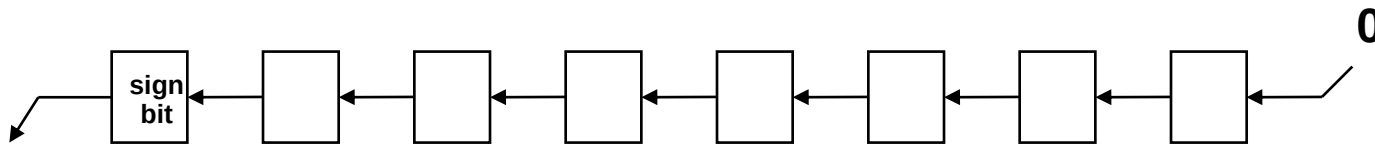


arithmetic shift right 1

Examples:

$110110 \rightarrow _11011 \rightarrow 111011$

- A left arithmetic shift operation:



arithmetic shift left 2

Examples:

$110110 \rightarrow 0110_ \rightarrow 011000$

ALU (Arithmetic Logic Unit)

- Performs all arithmetic and logic operations
- Key component of a computer's central processor unit.

Key Components of an ALU:

1.Arithmetic Unit: Responsible for performing arithmetic operations (addition, subtraction, etc.).

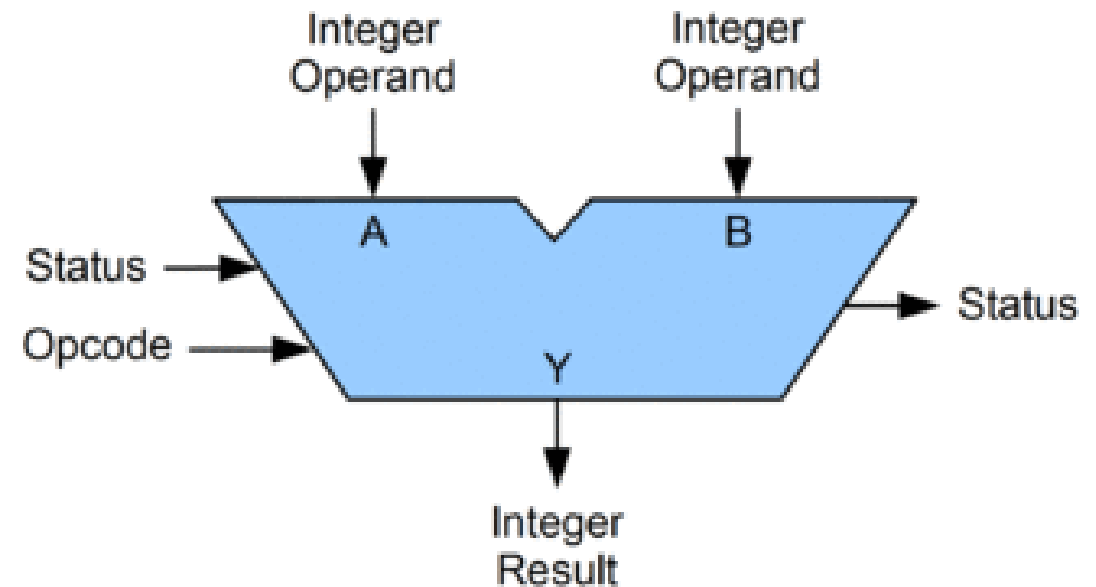
2.Logic Unit: Handles logical operations like AND, OR, XOR, and NOT.

3.Input Operands: The ALU typically receives two inputs (called operands) to operate on.

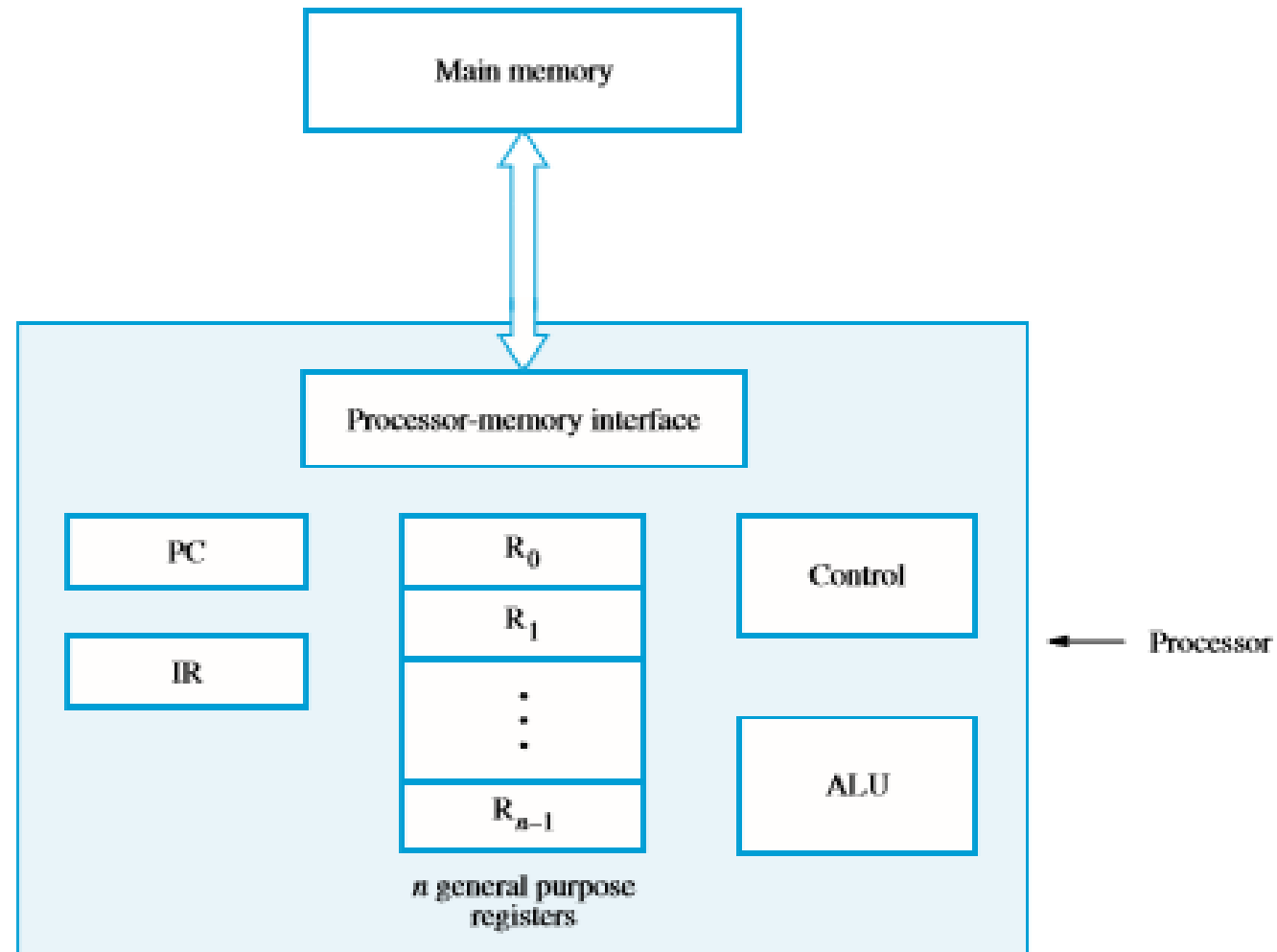
4.Control Signals/ Opcode: These determine which operation (arithmetic or logical) the ALU will perform.

5.Output: The result of the operation is sent as output to be used by other parts of the system.

6.Flags: Special outputs that indicate the status of the operation, such as overflow, zero result, or carry out.



Basic computer Operational Concepts



Operations Performed by the ALU:

1.Arithmetic Operations:

1. Addition
2. Subtraction
3. Multiplication (often through repetitive addition or dedicated circuits)
4. Division (often through repetitive subtraction or specialized logic)
5. Increment/Decrement

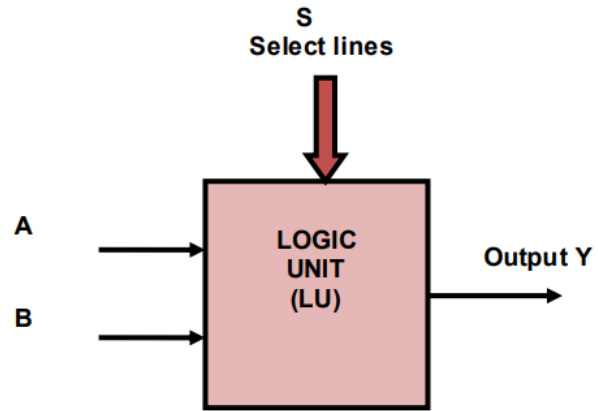
2.Logical Operations:

1. AND
2. OR
3. XOR (Exclusive OR)
4. NOT (Negation)

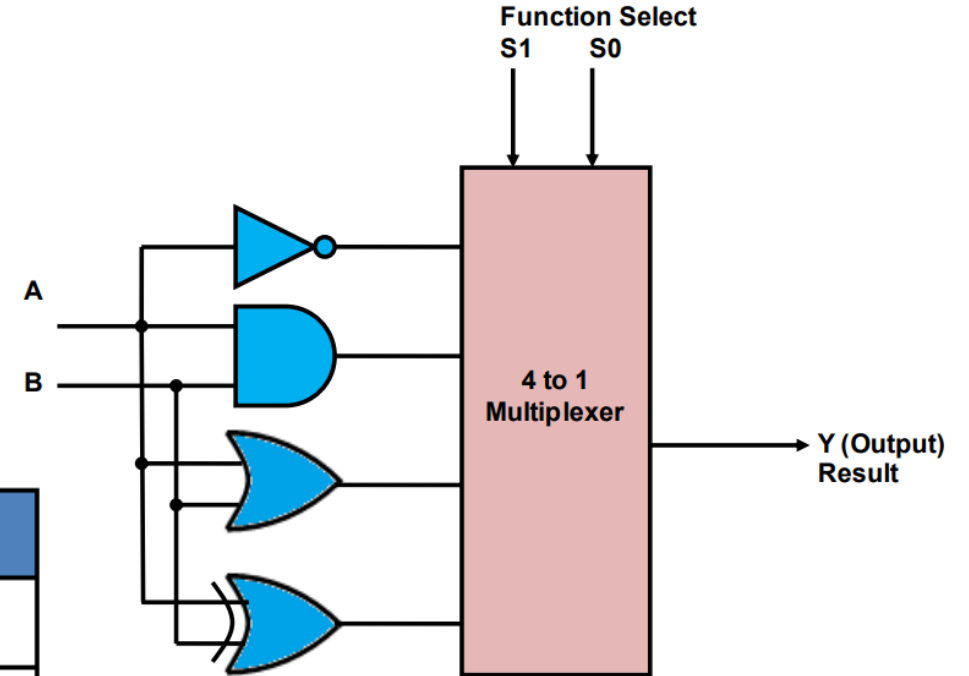
3.Bitwise Operations:

1. Shifting (left or right)
2. Rotation (left or right)

1-bit ALU design



S1	S0	Output	Operation
0	0	$Y=A'$	Complement
0	1	$Y=A \cdot B$	AND
1	0	$Y=A + B$	OR
1	1	$Y= A \oplus B$	XOR



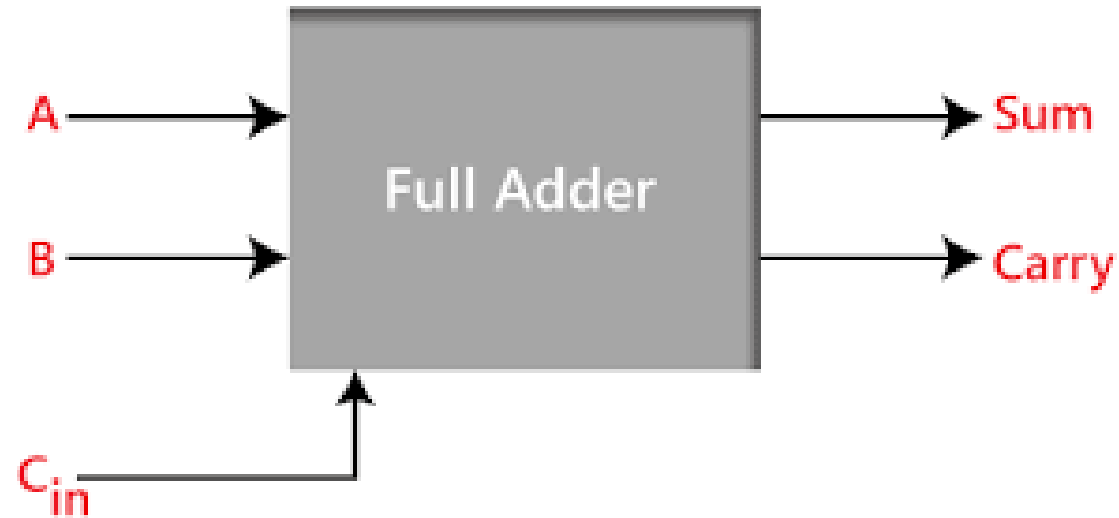
Design an arithmetic circuit with inputs A and B. Draw the logic diagram for the first two stages. The circuit generates the following four arithmetic operations (addition, subtraction, increment and decrement).

$$(A + B)$$

$$(A + 1)$$

$$(A - 1)$$

$$(A - B)$$



Design an arithmetic circuit with inputs A and B. Draw the logic diagram for the first two stages. The circuit generates the following four arithmetic operations (addition, subtraction, increment and decrement).

S	Cin	X	Y	
0	0	A	B	(A + B)
0	1	A	0	(A + 1)
1	0	A	1	(A - 1)
1	1	A	$\bar{B}$	(A - B)

